CHEAT SHEETS

High-Yield Pharmacology + Lab Value Shortcuts | NCLEX-RN Quick Review

How to use: Scan a table in 2-3 minutes, cover the right column, recall the nursing action, then practice a question. Clinical judgment matters more than memorizing a single number. Adult reference ranges are illustrative and vary by laboratory and context.

1 | Pharmacology suffix decoder

Pattern / clue	Common class or examples	High-yield nursing watch-out
-pril	ACE inhibitors: lisinopril, enalapril	Cough, angioedema, hyperkalemia; monitor BP, potassium, renal function. Avoid in pregnancy.
-sartan	ARBs: losartan, valsartan	Hyperkalemia, hypotension, renal changes; avoid in pregnancy.
-olol (many)	Beta blockers: metoprolol, propranolol	Bradycardia/hypotension; assess pulse and BP. Not every drug ending in -olol is interchangeable.
-dipine (many)	Dihydropyridine calcium-channel blockers: amlodipine	Peripheral edema, headache, hypotension.
-statin	Statins: atorvastatin, rosuvastatin	Report muscle pain/weakness or dark urine; review liver concerns and interactions.
-prazole	Proton pump inhibitors: omeprazole	Suppress gastric acid; monitor long-term adverse effects when applicable.
-xaban	Factor Xa inhibitors: apixaban, rivaroxaban	Bleeding risk; routine INR does NOT measure their anticoagulant effect.
-gliflozin	SGLT2 inhibitors: empagliflozin	Genital infections, volume depletion, euglycemic ketoacidosis risk.
-glutide	GLP-1 receptor agonists: semaglutide	GI adverse effects; evaluate severe persistent abdominal pain.
-cycline	Tetracycline antibiotics: doxycycline	Photosensitivity; separate from certain minerals/antacids; pregnancy/age cautions.

NCLEX trap: Suffixes are recognition clues, not proof of a drug class. Confirm the actual medication, indication, patient assessment, and order.

PHARMACOLOGY | Safety First

Fast associations: assess, anticipate, act

2 | High-alert medications & antidote associations

Medication / exposure	Key risk or monitoring	Reversal / immediate response clue
Insulin	Hypoglycemia; verify glucose, meal timing, type, and dose.	Treat hypoglycemia per protocol; severe cases may need IV dextrose or glucagon.
Unfractionated heparin	Bleeding; monitor aPTT or anti-Xa per protocol, platelets for HIT.	Protamine sulfate reverses UFH; urgent assessment for bleeding.
Warfarin	Bleeding; monitor INR, drug/diet interactions.	Vitamin K; major bleeding may require 4-factor PCC per protocol.
Opioids	Respiratory depression, sedation.	Support airway/ventilation; naloxone reverses opioid effects, may require repeat dosing.
Benzodiazepines	Sedation, respiratory compromise (especially with other depressants).	Support airway; flumazenil only in selected cases due to seizure/withdrawal risk.
Digoxin	Bradyarrhythmia, GI/visual symptoms; toxicity risk increases with low K.	Assess ECG, electrolytes, renal function; digoxin immune Fab for severe toxicity.
Magnesium sulfate	Loss of reflexes, respiratory depression, reduced urine output.	Stop infusion/assess; IV calcium gluconate may be used for severe toxicity per order.
Acetaminophen overdose	Hepatic injury; timing and dose matter.	N-acetylcysteine; urgent toxicology evaluation.
Potassium chloride IV	Rapid administration can cause fatal dysrhythmias.	Never IV push; use correct dilution/pump and facility safeguards.

3 | Class-to-clinical shortcut

Class	Remember this
Loop diuretics (furosemide)	Monitor K, BP, volume status, kidney function; risk of hypokalemia.
K-sparing diuretics (spironolactone)	Monitor K and renal function; risk of hyperkalemia.
Aminoglycosides (gentamicin)	Nephrotoxicity and ototoxicity; levels/renal monitoring per protocol.
Lithium	Narrow therapeutic index; dehydration, sodium changes, and interactions increase toxicity risk.
Corticosteroids	Hyperglycemia, infection risk; tapering depends on dose and duration.
Antipsychotics	Watch for extrapyramidal symptoms, NMS, metabolic effects; some prolong QT.

PHARMACOLOGY | NCLEX Traps

Recognize the assessment that changes the plan

4 | Five quick priority rules

Cue	First nursing thought
New facial/lip/tongue swelling on ACE inhibitor	Possible angioedema: airway assessment and urgent escalation.
Insulin order + patient is NPO	Verify insulin type, glucose, and protocol; do not assume every insulin dose is withheld.
Heparin + platelet drop / new thrombosis	Suspect HIT; stop heparin and escalate per protocol.
Opioid + slow respirations / decreased responsiveness	Airway and ventilation first; naloxone when indicated.
Digoxin + nausea + bradycardia + low potassium	Possible toxicity; assess, hold pending guidance, notify prescriber.
Diuretic + weakness / palpitations	Think electrolyte loss or excess; assess rhythm and potassium.
Anticoagulant + black stool / severe headache	Potential serious bleeding; immediate assessment and escalation.
Antibiotic + wheeze / hypotension / hives	Possible anaphylaxis; emergency response, epinephrine per protocol.

5 | Medication administration mini-check

Before: allergies, indication, patient identity, dose/route/time, vitals, relevant labs, contraindications and interactions.
During: use independent double-checks when policy requires, correct pump settings, and monitor for reaction.
After: reassess response, document, teach, and escalate unexpected changes.

6 | Memory hooks

Hook	Meaning
ACE = Airway, Creatinine, Electrolytes	ACE inhibitor: angioedema, renal function, potassium.
Loops lose K	Loop diuretics often lower potassium.
Spironolactone spares K	Potassium-sparing means hyperkalemia risk.
Opioid overdose = breathe first	Ventilation is immediate priority; naloxone is an adjunct.
Anticoagulants = bleeding surveillance	Look beyond the lab: assess the patient for bleeding.

LAB VALUES | Core Adult Shortcuts

Use the reference interval printed by the actual laboratory.

7 | Electrolytes, kidney function & glucose

Test	Illustrative adult range	Why it matters / NCLEX cue
Sodium (Na)	135-145 mEq/L	Neurologic changes; rapid shifts can be dangerous.
Potassium (K)	3.7-5.2 mEq/L	High or low K can cause life-threatening dysrhythmias.
Chloride (Cl)	96-106 mEq/L	Interpret with sodium and acid-base balance.
Total CO2 / bicarbonate	23-29 mEq/L (chemistry)	Low often suggests metabolic acidosis; correlate with ABG.
Total calcium	8.5-10.2 mg/dL	Neuromuscular changes; albumin affects total calcium.
BUN	6-20 mg/dL	Hydration, renal perfusion and kidney context.
Creatinine	Approx. 0.6-1.3 mg/dL	Renal function; varies with muscle mass and sex.
Fasting glucose	70-100 mg/dL (typical)	<70 mg/dL is a common hypoglycemia alert threshold.
Albumin	3.4-5.4 g/dL	Nutrition/inflammation/liver context; affects total calcium.

8 | ABG interpretation: ROME

Parameter	Typical arterial range	Interpretation shortcut
pH	7.35-7.45	<7.35 acidemia; >7.45 alkalemia.
PaCO2	35-45 mmHg	Respiratory: CO2 moves opposite pH (ROME).
HCO3-	22-26 mEq/L (ABG)	Metabolic: bicarbonate moves with pH.
PaO2	80-100 mmHg (sea level)	Oxygenation; varies with age, altitude, oxygen support.

Shortcut: Low pH + high CO2 = respiratory acidosis. Low pH + low HCO3- = metabolic acidosis. Always assess compensation and clinical context.

LAB VALUES | Hematology + Coagulation

Numbers are clues. Symptoms and trends determine urgency.

9 | CBC and coagulation

Test	Common adult reference / target	High-yield interpretation
WBC	~4,500-11,000 /microL	Infection, inflammation, marrow suppression; neutrophil count matters.
Hemoglobin	~12-16 g/dL (women); ~13.5-17.5 (men)	Oxygen-carrying capacity; assess bleeding and symptoms.
Platelets	150,000-450,000 /microL	Low counts increase bleeding risk; HIT possible with heparin.
INR (not anticoagulated)	~0.8-1.2	Warfarin target often 2-3, but depends on indication.
aPTT	~25-35 sec (lab dependent)	UFH targets are institution/assay-specific; not a universal fixed number.

10 | What is urgent? Think patient + trend

Finding	Priority assessment
K abnormal + palpitations/ECG changes	Cardiac monitoring and immediate escalation.
Low glucose + confusion/diaphoresis	Check glucose; treat promptly per hypoglycemia protocol.
Low hemoglobin + hypotension/tachycardia	Assess active bleeding and perfusion.
Low platelets + new bleeding	Bleeding precautions and urgent clinical review.
Creatinine rising + reduced urine output	Evaluate perfusion, medications, and acute kidney injury.
Sodium abnormal + seizures/confusion	Neurologic safety, urgent escalation, careful correction.

11 | Quick self-test

1. A patient taking spironolactone has K 5.8 mEq/L. What is your first assessment?
2. A patient receiving opioids is difficult to arouse and breathing 6/min. What takes priority?
3. pH 7.29, PaCO2 55, HCO3- 25: Which primary acid-base disorder?

Answers: 1. Assess symptoms/rhythm and escalate hyperkalemia. 2. Airway and ventilation; emergency response/naloxone as indicated. 3. Respiratory acidosis.

QUICK REFERENCE | Safety & Sources

PrepRX | Your Prescription for NCLEX Success

12 | 60-second pre-exam recall

Think	Ask yourself
Airway + breathing	Is the patient ventilating and oxygenating?
Perfusion	Are BP, mentation, urine output and rhythm stable?
Medication risk	Which lab or vital sign changes whether this medication is safe?
Trend > isolated number	Is the result new, rapidly changing, or symptomatic?
Clinical judgment	What cue is most concerning, and what action reduces immediate harm?

Important educational note

These CHEAT SHEETS are educational study aids, not medication orders or bedside protocols. Reference ranges differ across laboratories, age groups, pregnancy, altitude, assays, and clinical conditions. Drug reversal and dosing decisions require current prescribing information, clinical evaluation, and institutional protocols. Do not use suffixes alone to identify medications.

Reference foundation

NCSBN: 2026 NCLEX-RN Test Plan (effective April 2026) - pharmacological/parenteral therapies and clinical judgment. ncsbn.org/publications/2026-nclex-rn-test-plan

MedlinePlus / U.S. National Library of Medicine: Comprehensive metabolic panel; Basic metabolic panel; Sodium blood test; Potassium test; Creatinine blood test. medlineplus.gov

Medication safety: Verify drug-specific risks, antidotes, and monitoring against current FDA-approved prescribing information and local protocols. accessdata.fda.gov/scripts/cder/daf/

PrepRX Study Strategy: Pair every shortcut with a patient scenario: recognize cues, analyze cues, prioritize hypotheses, generate solutions, take action, and evaluate outcomes.